

XAI for U: Explainable AI for Ubiquitous, Pervasive and Wearable Computing

Martin Gjoreski
martin.gjoreski@usi.ch
Università della Svizzera italiana
Lugano, Switzerland

Teena Hassan
teena.hassan@h-brs.de
Bonn-Rhein-Sieg University of
Applied Sciences
Sankt Augustin, Germany

Mor Vered
mor.vered@monash.edu
Monash University
Melbourne, Australia

Sebastian Houben
sebastian.houben@h-brs.de
Bonn-Rhein-Sieg University of
Applied Sciences
Sankt Augustin, Germany

Stefan Kopp
skopp@techfak.uni-bielefeld.de
Bielefeld University
Bielefeld, Germany

ABSTRACT

The workshop *XAI for U* aims to address the critical need for transparency in Artificial Intelligence (AI) systems that integrate into our daily lives through mobile systems, wearables, and smart environments. Despite advances in AI, many of these systems remain opaque, making it difficult for users, developers, and stakeholders to verify their reliability and correctness. This workshop addresses the pressing need for enabling Explainable AI (XAI) tools within Ubiquitous and Wearable Computing and highlights the unique challenges that come with it, such as XAI that deals with time-series and multimodal data, XAI that explains interconnected machine learning (ML) components, and XAI that provides user-centered explanations. The workshop aims to foster collaboration among researchers in related domains, share recent advancements, address open challenges, and propose future research directions to improve the applicability and development of XAI in Ubiquitous Pervasive and Wearable Computing — and with that seeks to enhance user trust, understanding, interaction, and adoption, ensuring that AI-driven solutions are not only more explainable but also more aligned with ethical standards and user expectations.

CCS CONCEPTS

• **Human-centered computing** → *Ubiquitous and mobile computing*; • **Computing methodologies** → *Artificial intelligence*.

KEYWORDS

XAI, explainable AI, Ubiquitous Wearable Computing, Wearable Computing, User Studies

ACM Reference Format:

Martin Gjoreski, Teena Hassan, Mor Vered, Sebastian Houben, and Stefan Kopp. 2024. XAI for U: Explainable AI for Ubiquitous, Pervasive and Wearable Computing. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion*

'24), October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3675094.3677571>

1 BACKGROUND AND MOTIVATION

1.1 Introduction to XAI for Ubiquitous Computing

Ubiquitous and Wearable Computing, exemplified by mobile systems, wearable technologies, and intelligent environments, utilizes Artificial Intelligence (AI) to develop advanced solutions that improve many aspects of our lives, from healthcare and sports performance to general quality of life. Furthermore, ubiquitous technologies allow users to interact with AI-enabled agents through a diverse array of devices and everyday objects, such as computers, smartphones, smart glasses, smart mattresses, and refrigerators. In the age of ubiquitous AI-based systems that deeply integrate into human lives, it is critical that users understand how these systems operate and make decisions. Unfortunately, today's groundbreaking AI methods are black-boxes (i.e., the decision model and the decision process are not understandable). This opacity unveils a critical need for Explainable AI (XAI) tools, through which all stakeholders (e.g. users, developers, employers) can verify the correctness and reliability of decisions made by AI-driven ubiquitous solutions. As AI continues to integrate into our lives, making its operations transparent through XAI is not just a preference but also a necessity.

The development of XAI began in the 1980s with systems like XPLAIN and EMYCIN [15, 16], which aimed to explain expert systems. By the late 1990s, the focus expanded to explainers for Bayesian networks [1, 9] and recommender systems [5, 7]. In recent years, the need for XAI has grown with the rise of complex AI approaches, which are continuously applied also in the context of Ubiquitous and Wearable Computing. For example, studies have explored XAI for Human Activity Recognition models [3, 4, 8]; Sharma et al. [14] utilized XAI in their machine learning (ML) solution for psychophysiological arousal in children who stutter; Rahman et al. [12] designed an interpretable framework to empower people with Parkinson's disease [12]; Das Antar et al. explored quantitative and qualitative model explanations to understand models to predict physical functioning of people with Multiple Sclerosis [2]. Most XAI approaches are aimed at generating post-hoc explanations for



This work is licensed under a Creative Commons Attribution International 4.0 License.

UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1058-2/24/10
<https://doi.org/10.1145/3675094.3677571>

single decisions or actions of an AI-based system. Current work extends this to treating explanations as an interactive process though which an explanation (considered as successful understanding of the explanandum) is co-constructed by a machine explainer and a human explainee [13].

In addition to the increasing trend of XAI studies, UbiComp studies have shown that the availability of explanations positively influenced the willingness of users to accept personal sensing technology [10]. Furthermore, two studies have shown that end users do not even understand a high-level overview of Fitbit and similar sensing platforms [6, 17]. However, despite the clear need for XAI at UbiComp, there has not been a dedicated XAI workshop at UbiComp so far.

1.2 Why should XAI for U be a part of UbiComp/ISWC 2024

XAI has become a particularly hot topic in recent years: the European Union engrossed the AI Act, which mandates human oversight, and increased tasks in which XAI can help [11]. DARPA launched the XAI initiative; HORIZON Europe continuously holds XAI-related project calls, The Swiss Hasler foundation opened the Responsible AI Funding Program, etc. Additionally, major AI conferences, such as IJCAI 2023, AAAI 2024, and NeurIPS 2023, have featured workshops dedicated to XAI as well as dedicated tracks, indicating its significance in the research community. Since 2023, there is also an international conference dedicated to XAI. While these venues emphasize algorithmic advancements, the unique challenges of applying XAI within the context of Ubiquitous and Wearable Computing are not adequately addressed. Challenges include:

- (1) The application of XAI has been largely focused on image and text modalities, with less attention on time-series data from wearable sensors and IoT devices, which are central to Ubiquitous and Wearable Computing.
- (2) There is a tendency in XAI research to focus on models utilizing a single data source, neglecting the complex, multimodal data prevalent in Ubiquitous and Wearable Computing.
- (3) Existing XAI tools often explain only a single model within a system, ignoring the interconnectedness of multiple ML components used in Ubiquitous and Wearable Computing.
- (4) Conducting user studies to evaluate XAI tools poses significant challenges, leading to a scarcity of end-user evaluations in the XAI literature.
- (5) Many XAI solutions are static and do not allow for adjustments in input and output variables, rendering the static XAI solutions unapplicable in the dynamic field of Ubiquitous and Wearable Computing.
- (6) A general lack of user understanding regarding the operations of platforms like Fitbit, indicating a need for end-to-end XAI solutions that clarify the entire sensing system and interactions between various ML components.
- (7) Privacy and ethical considerations, which have always been central to Ubiquitous and Wearable Computing, necessitate a careful integration of XAI tools within these systems.

- (8) The deployment of XAI on wearable devices remains under-explored, requiring focused attention to improve user trust and understanding.
- (9) The diversity in explanation needs of end users necessitate XAI solutions in Ubiquitous and Wearable Computing that can support personalised explanations at different levels of granularity.
- (10) The use of XAI to generate actionable recommendations to end-users of Ubiquitous and Wearable Computing demands ease of adaptability to varying contexts of use.
- (11) The effects of explanations on users' over and under-reliance on decision support tools and how to create an appropriate level of trust have not been investigated broadly and would enable users to effectively use and engage with the explanations rather than follow them blindly.

2 OBJECTIVES

The objective of *XAI for U* Workshop is to improve the applicability and development of explainable AI within the use of Ubiquitous and Wearable computing. This goal is achieved through a series of structured activities:

- (1) Bringing together researchers from the field: The workshop facilitates collaboration and exchange of ideas among experts in XAI and Ubiquitous and Wearable Computing, providing a platform for networking and the establishment of potential collaborations.
- (2) Discussing recent developments: The workshop offers an opportunity to share the latest research findings, technological advancements, and methodological approaches in XAI, helping to keep the community up-to-date.
- (3) Discussing open/insufficiently addressed challenges: It identifies and deliberates on the current gaps, limitations, and unsolved problems in XAI applications to Ubiquitous and Wearable Computing, encouraging a focus on pressing issues that need attention.
- (4) Proposing action items for future research: The event aims to generate a roadmap or set of recommendations for addressing identified challenges, guiding future research directions and efforts in the field.

3 WORKSHOP STRUCTURE

The workshop includes five main segments, including two keynote presentations, an oral presentations, a poster session, a panel discussion, and group discussions. The keynote presentations will be delivered by Prof. Tim Miller (University of Queensland) and Prof. Brian Lim (National University of Singapore) who have confirmed their willingness to contribute. Following each keynote, the agenda includes three to five short oral presentations by authors of selected workshop papers. These authors will also be encouraged to prepare posters to facilitate deeper discussions and networking. In an effort to broaden the impact of the workshop, we will invite authors of papers related to XAI from the main conference to participate at least in the poster session, i.e., these authors will also be invited to prepare posters for our workshop, thus enriching the variety and depth of content available.

The panel discussion will include 4 to 5 experts in the field, including the two keynote speakers. Group discussions will form another crucial part of the workshop, with the intention of dividing participants into 3 to 4 groups. Each group will focus on distinct open challenges in XAI for Ubiquitous and Wearable Computing, brainstorming possible approaches and solutions. In the end, each group will give a short presentation to share their findings.

The workshop will conclude with a summary of the day's discussions and a farewell, encapsulating the insights gained and ideas generated.

Tentative schedule:

- 09:00 - 09:15: Welcome
- 09:15 - 10:00: Keynote 1 + Q&A
- 10:00 - 10:15: Coffee break + Poster tour
- 10:15 - 11:00: Oral presentations – Part 1
- 11:00 - 11:45: Keynote 2 + Q&A
- 11:45 - 13:15: Lunch break
- 13:15 - 14:00: Oral presentations – Part 2
- 14:00 - 15:00: Panel discussion
- 15:00 - 15:15: Coffee break + Poster tour
- 15:15 - 16:15: Group discussion (3-4 groups)
- 16:15 - 17:00: Group discussion – summary
- 17:00 - 17:30: Conclusion and farewell

4 WORKSHOP SUBMISSIONS

We invite submissions of original research, insightful case studies, and work in progress exploring XAI for Ubiquitous and Wearable Computing approaches. Submissions should be short, up to 4 pages, including references. Our review process will be brief and double-blind through the Precision Conference System (PCS). The accepted papers will be published in the ACM Digital Library and the Adjunct Proceedings. We anticipate receiving around 15 - 20 paper submissions and hosting 25 - 30 participants. The workshop will be open to both registered participants and those who have not registered explicitly for our workshop.

The pre-workshop preparation includes a plan to engage the community and attract quality submissions, including setting up a dedicated workshop website¹, issuing a call for papers, and leveraging social media platforms and mailing lists for widespread dissemination. We will also engage directly with potential attendees through personalized invitations to build momentum and ensure a successful and well-attended workshop.

5 IMPACT AND LONG-TERM OBJECTIVES

The workshop is expected to have a significant impact on the community by fostering a focused discussion on XAI within Ubiquitous and Wearable Computing. Given the absence of similar workshops in recent years at UbiComp, this workshop will encourage the exchange of ideas, advancements, and solutions specific to XAI for Ubiquitous and Wearable Computing. It has the potential to catalyze a series of workshops at subsequent UbiComp conferences, thus nurturing a dedicated community of researchers and practitioners committed to advancing XAI in Ubiquitous and Wearable Computing.

¹<https://ubicom-p-xai.github.io>

As a post-workshop activity, a comprehensive summary report or whitepaper will be released (e.g., on arxiv and shared via email with all workshop participants), encapsulating the discussions, insights, and conclusions from the workshop. This document will serve as a valuable resource for the wider research community, offering a synthesized overview of the state-of-the-art and future pathways for XAI in Ubiquitous and Wearable Computing. We will also consider publishing the workshop summary (as a peer-reviewed publication) and even using it as the basis for a scoping review.

6 ORGANIZERS' BACKGROUNDS

Martin Gjoreski an Ambizione Fellow funded by the Swiss National Science Foundation (SNSF). His primary research focus is on AI, specifically in the areas of ML, federated learning, and XAI. He applies these concepts to various domains such as Ubiquitous Computing, Affective Computing, and Digital Healthcare. He co-organized the UbiTention Workshop at UbiComp 2020 and is an Associate Editor at IMWUT since 2023.

Teena Hassan is a Professor for Computer Science and the Director of the Institute for Artificial Intelligence and Autonomous Systems (A2S) at the Bonn-Rhein-Sieg University of Applied Sciences in Germany. She specialises in affective computing and human-robot interaction for diverse assistive scenarios, especially in the healthcare sector. Her research also includes application of XAI methods to pain recognition and development of software architecture for autonomous explanation generation for social robots. She co-organised a workshop on empathic human-robot interaction at the European Robotics Forum 2017 and will be co-organising a workshop titled "Multimodal Co-Construction of Explanations with XAI" at ICM 2024.

Mor Vered is a senior Lecturer at the Dept of DS & AI at Monash University. Her research interests lie in the interaction between humans and intelligent agents, where she work to incorporate lessons and inspirations from cognitive science, neuroscience and biology. As such she specialises in human-centered XAI, generating explanations built on cognitive theories, considering that these explanations need to be consumed by people and therefore taking into account human situation awareness models. She organized the AAMAS-24 Workshop on Rebellious and Disobedient Agents in AI (RaD-AI) and is co-organiser of the "Explainable AI workshop" at IJCAI, 2024.

Sebastian Houben is a Professor for Robot Vision and Machine Learning at the Bonn-Rhein-Sieg University of Applied Sciences and team lead at the nearby Fraunhofer Institute for Intelligent Analysis and Information Systems. His research covers data-driven methods of sensor processing, in particular utilizing multi-modal foundation models for application-driven robotics as well as data-driven simultaneous localization and mapping. He organized the "German Traffic Sign Detection Workshop" at IJCNN in 2013.

Stefan Kopp is Professor for Cognitive and Socially Interactive Systems and head of the Social Cognitive Systems Group at Bielefeld University, Germany. He is currently co-coordinator of the research center on "Cognitive Interaction Technology" (CITEC) and member of a collaborative research center on "Constructing Explainability" (CRC 318). His research interests are centered around the

cognitive and interactive mechanisms of social interaction, communication and cooperation, and how such skills can be transferred to intelligent technical systems to enable new levels of human-technology cooperation. Stefan has served as conference (co-)chair, program chair, or senior program committee member at several international conferences or workshops in the field of interactive intelligent systems (e.g. AAMAS, IVA, HRI, ICMI, CogSci). He will be co-organising a workshop on “Multimodal Co-Construction of Explanations with XAI” at ICMI 2024.

ACKNOWLEDGMENTS

We acknowledge the financial support for this workshop from:

- Swiss National Science Foundation (SNSF), Project XAI-PAC: Towards Explainable and Private Affective Computing (PZ00P2_216405).
- Ministerium für Kultur und Wissenschaft (MKW) des Landes Nordrhein-Westfalen (NRW), “Profilbildung 2022” project: Zentrum Assistive Technologien Rhein-Ruhr.
- Deutsche Forschungsgemeinschaft (DFG): TRR 318/1 2021 – 438445824 (<https://gepris.dfg.de/gepris/projekt/438445824>).

REFERENCES

- [1] Robert Andrews, Joachim Diederich, and Alan B Tickle. 1995. Survey and critique of techniques for extracting rules from trained artificial neural networks. *Knowledge-based systems* 8, 6 (1995), 373–389.
- [2] Anindya Das Antar, Anna Kratz, and Nikola Banovic. 2023. Behavior Modeling Approach for Forecasting Physical Functioning of People with Multiple Sclerosis. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 1 (2023), 1–29.
- [3] Luca Arrotta, Gabriele Civitarese, and Claudio Bettini. 2022. Dexar: Deep explainable sensor-based activity recognition in smart-home environments. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 1 (2022), 1–30.
- [4] Luca Arrotta, Gabriele Civitarese, and Claudio Bettini. 2024. Semantic Loss: A New Neuro-Symbolic Approach for Context-Aware Human Activity Recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 4 (2024), 1–29.
- [5] Henriette Cramer, Vanessa Evers, Satyan Ramlal, Maarten Van Someren, Lloyd Rutledge, Natalia Stash, Lora Aroyo, and Bob Wielinga. 2008. The effects of transparency on trust in and acceptance of a content-based art recommender. *User Modeling and User-adapted interaction* 18 (2008), 455–496.
- [6] Rachel Eardley, Sue Mackinnon, Emma L Tonkin, Ewan Soubutts, Amid Ayobi, Jess Linington, Gregory JL Tourte, Zoe Banks Gross, David J Bailey, Russell Knights, et al. 2022. A case study investigating a user-centred and expert informed ‘companion guide’ for a complex sensor-based platform. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2 (2022), 1–23.
- [7] Jonathan L Herlocker, Joseph A Konstan, and John Riedl. 2000. Explaining collaborative filtering recommendations. In *Proceedings of the ACM Conference on Computer Supported Cooperative Work*. 241–250.
- [8] Jeya Vikranth Jeyakumar, Ankur Sarker, Luis Antonio Garcia, and Mani Srivastava. 2023. X-CHAR: A Concept-based Explainable Complex Human Activity Recognition Model. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 1 (2023), 1–28.
- [9] Carmen Lacave and Francisco J Diez. 2002. A review of explanation methods for Bayesian networks. *The Knowledge Engineering Review* 17, 2 (2002), 107–127.
- [10] Joshua Newn, Ryan M Kelly, Simon D Alfonso, and Reeva Lederman. 2022. Examining and Promoting Explainable Recommendations for Personal Sensing Technology Acceptance. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 3 (2022), 1–27.
- [11] Cecilia Panigutti, Ronan Hamon, Isabelle Hupont, David Fernandez Llorca, Delia Fano Yela, Henrik Junklewitz, Salvatore Scalzo, Gabriele Mazzini, Ignacio Sanchez, Josep Soler Garrido, et al. 2023. The role of explainable AI in the context of the AI Act. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. 1139–1150.
- [12] Wasifur Rahman, Abdelrahman Abdelkader, Sangwu Lee, Phillip Yang, Md Saiful Islam, Tariq Adnan, Masum Hasan, Ellen Wagner, Sooyong Park, E Ray Dorsey, et al. 2024. A User-Centered Framework to Empower People with Parkinson’s Disease. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 4 (2024), 1–29.
- [13] Katharina J Rohlfing, Philipp Cimiano, Ingrid Scharlau, Tobias Matzner, Heike M Buhl, Hendrik Buschmeier, Elena Esposito, Angela Grimminger, Barbara Hammer, Reinhold Häb-Umbach, et al. 2020. Explanation as a social practice: Toward a conceptual framework for the social design of AI systems. *IEEE Transactions on Cognitive and Developmental Systems* 13, 3 (2020), 717–728.
- [14] Harshit Sharma, Yi Xiao, Victoria Tumanova, and Asif Salekin. 2022. Psychophysiological Arousal in Young Children Who Stutter: An Interpretable AI Approach. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 3 (2022), 1–32.
- [15] William R Swartout. 1983. XPLAIN: A system for creating and explaining expert consulting programs. *Artificial Intelligence* 21, 3 (1983), 285–325.
- [16] William Van Melle, Edward H Shortliffe, and Bruce G Buchanan. 1984. EMYCIN: A knowledge engineer’s tool for constructing rule-based expert systems. *Rule-based expert systems: The MYCIN experiments of the Stanford Heuristic Programming Project* (1984), 302–313.
- [17] Lev Velykoivanenko, Kavous Salehzadeh Niksirat, Noé Zufferey, Mathias Humbert, Kevin Huguenin, and Mauro Cherubini. 2021. Are those steps worth your privacy? Fitness-tracker users’ perceptions of privacy and utility. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 4 (2021), 1–41.